Effect of K and Mn Promoters on Surface Intermediates Formation over $Fe/YZrO_x$ Catalysts under CO_2 Fischer-Tropsch Synthesis

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Introduction

The CO_2 Fischer-Tropsch synthesis (CO_2-FTS) of C_2-C_4 olefins and C_{5+}-hydrocarbons is a promising approach to convert this greenhouse gas. Iron-based catalysts, in both supported and bulk forms, have been widely used for this purpose. It is well known that alkali and transition metals used as promoters increase the selectivity to the target products.[1,2] However, their influence and role in this reaction are still not well understood. In this respect, *Operando* DRIFT spectroscopy at high-temperatures and pressures can deliver useful information about surface intermediates and their transformations. Against this background, this technique in time-resolved and stationary modes was combined with steady-state catalytic tests to derive mechanistic insights, including surface intermediates, into the role of potassium and manganese promoters for $Fe/YZrO_x$ in CO_2-FTS.

Experimental

A series of promoted $Fe/YZrO_x$ catalysts was prepared by incipient wetness impregnation of $YZrO_x$ with $Fe(NO_3)_3$*$9H_2O$, KNO_3, $Mn(NO_3)_2$*$4H_2O$ followed by drying and calcination at 500°C. The *operando* DRIFTS experiments were performed on a Nicolet iS10 spectrometer equipped with a high-pressure/high-temperature reaction DRIFTS cell (Harrick) connected to a gas supplying system and to an online mass spectrometer (OmniStar). After pretreatment (H_2/He, 1:1, 400°C, 120 min) the catalyst was exposed to the CO_2 (22 vol.% in He) for 60 min followed by exposure to H_2 (10 vol.% in He) or to the reaction gas mixture (CO_2/H_2, 1/3, 15 bar) at 300°C and 15 bar.

Results and Discussion

The DRIFT spectra of $Fe/YZrO_x$ catalysts without or with K and/or Mn were recorded at 300°C and 15 bar after exposure to CO_2 followed by feeding H_2 (Fig.1). The surface of the $Fe/YZrO_x$ and Mn-$Fe/YZrO_x$ catalysts after CO_2 adsorption were only characterized by carbonate species with different structures (1600 – 1300 cm^{-1}).[3] When the catalysts were hereafter treated in a flow of H_2 these species disappeared either by desorption of CO_2 or reduction by H_2 to gaseous CO.

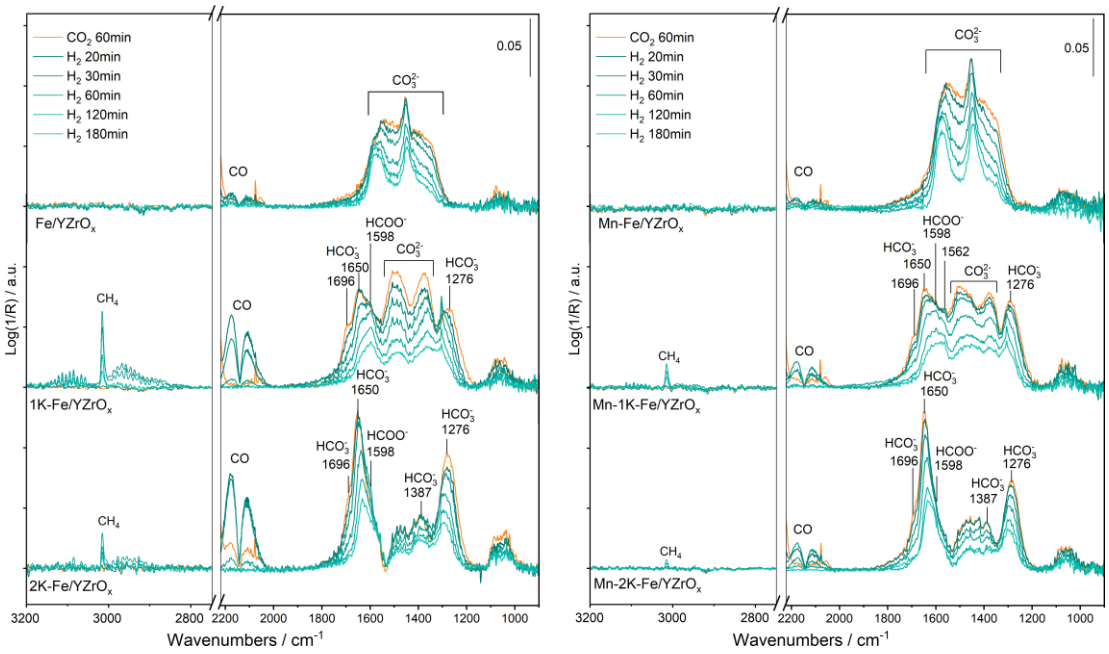

Figure 1. DRIFT spectra of Fe/YZrO$_x$ catalyst promoted with K and Mn measured during subsequent exposure to CO_2 and H_2 (10 vol.% in He) treatment at 300°C and 15 bar.

The introduction of K resulted in the formation of additional bicarbonate species after catalyst treatment in CO_2 (1696, 1650, 1387, 1276 cm^{-1}). Formate intermediates ($HCOO^-$) were detected after treatment in H_2 and converted into gaseous CO and methane. The broad peak in the 3000-2900 cm^{-1} region especially noticeable in the spectra of the 1K-Fe/YZrO$_x$ catalyst indicates the formation of higher hydrocarbons. It is worth mentioning that all Mn-promoted catalysts showed similar surface adsorbates like the unpromoted analogs, but lower amounts of CO and methane were formed during H_2 exposure. Thus, the Mn promotion does not have a significant effect on adsorbates, but it does affect the dynamics of their transformation after switching CO_2 to H_2 atmosphere.

The obtained DRIFTS data showed that K promotes the formation of additional surface adsorbates and intermediates, while the Mn promoter might stabilize these adsorbates and/or blocks H_2 adsorption sites for their further hydrogenation.

References

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